

Lithium-Ion vs. Novel Zinc-Based Gel Polymer Batteries: Comparing Energy Density, Capacity Retention, and Cost

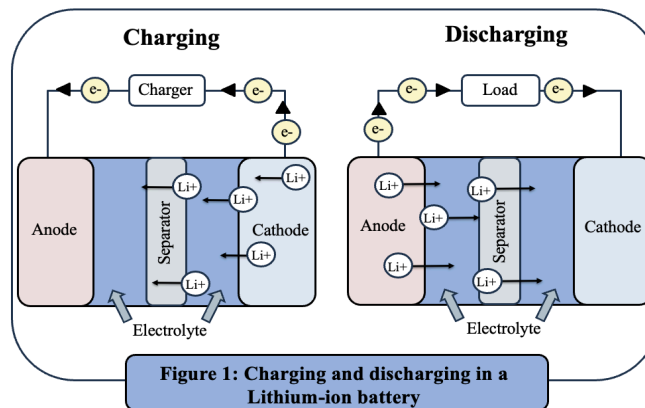
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Abstract

As society transitions toward cleaner energy sources, the ability to store intermittent power becomes essential, placing renewed scientific focus on advancing battery technology. Currently, commercialized lithium-ion batteries present several significant disadvantages. Severe dendritic growth shortens the batteries' lifespan, as these structures pierce the separator and cause short circuits, intense heat, and even potential fires. Furthermore, lithium-ion batteries often include toxic liquid electrolytes that can leak during disposal. This not only harms the environment, but it also is detrimental to human health. However, due to the lithium-ion batteries' high energy density, they have risen in popularity. Still, many of the disadvantages of these batteries are overlooked, and through design modifications, these issues can be mitigated and avoided completely. For example, implementing polymer electrolytes rather than liquid electrolytes can effectively prevent dendrite growth, as the polymer's strength limits dendrite penetration. The polymer electrolyte is also non-flammable, preventing fires, and due to its gel-like behavior, it prevents toxic leakage. If the separator could be replaced with the polymer electrolyte, the battery would no longer have to be stiff, allowing it to become flexible, therefore offering additional resilience and functionality even after puncture. Moreover, replacing lithium with zinc could offset some of the environmental issues. My project will explore and compare the energy efficiency, cost, and cycle retention of the zinc-based battery with modifications with the standard lithium-ion batteries using a multimeter. I will modify the zinc-based polymer electrolyte battery until it achieves a higher efficiency in each aspect of the comparison.

Background

Lithium-ion batteries consist of a few parts: the anode, cathode, separator, and electrolytes. Discharging is when a battery is actively being used. Charging is when a battery is being charged. When discharging, electrons flow from the anode to the load to the cathode while the lithium ions travel from the anode to the separator to the cathode. When charging, electrons flow from the cathode to the charger to the anode while, and the lithium ions travel from the cathode to the separator to the anode (see Figure 1 for diagram). One full cycle in a battery is when it completely charges or discharges. However, the energy capacity after each cycle decreases due to the degradation and loss of materials in the chemical reactions. After a certain amount of cycles, the battery is depleted, and the term cycle retention is the ability to maintain energy capacity over these cycles.



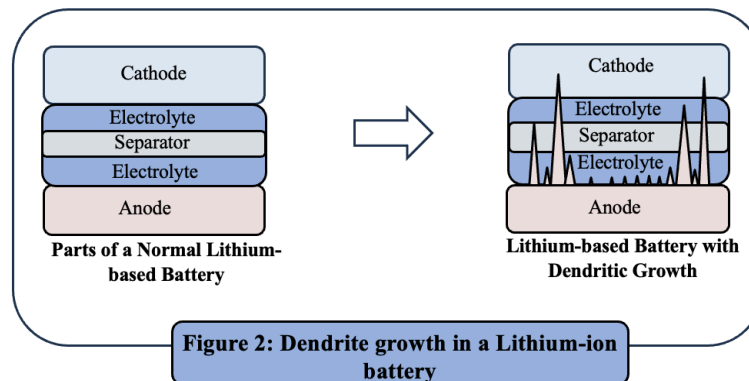
Problem

Lithium-ion batteries are rising in prevalence due to their energy density, which is the amount of energy capacity per mass or volume. Yet many issues are overlooked. Firstly, there are many environmental issues that come with the lithium-ion battery. The extraction of lithium is extremely detrimental, as it causes heavy water depletion, land degradation, carbon dioxide emissions, and ecosystem damage. Also, most of these batteries contain liquid electrolytes,

which could leak, resulting in pollution and health failure. The harmful materials in these batteries don't have strict disposal regulations, meaning they could severely harm the environment (Kang et al., 2013). Secondly, there are issues with lithium-ion batteries and their dendrite growth (see Figure 2 for diagram of dendrite growth). Dendrite growth, which are needle-like structures that can pierce the separator, can lead to short circuits, which could completely destroy the battery. It could also result in intense heat and fires, and dendrite growths are a major reason for car fires. This issue can become a major hazard and safety issue. Finally, lithium-ion batteries aren't super resilient, and when damage is inflicted, they often stop working or their performance diminishes. Also, my research tries to improve retention of energy in each cycle and cost efficiency.

Current Work

Current works have identified the many challenges previously stated. However, many



studies target just one of these issues in isolation: environmental impact, dendrite growth, energy efficiency, and battery resilience. For example, the study *Lithium-ion batteries and the future of sustainable energy: A comprehensive review* targets the environmental and safety issues from a lithium-ion battery and gives modifications, yet it handles these issues individually and doesn't give an overall analysis of the energy density, cycle retention, and cost of the altered battery (Ngoy et al., 2025). In contrast, this project aims to give an overall comprehensive analysis of

both batteries to truly understand if the novel zinc-based battery could be comparable to the standardized lithium-ion battery.

Solution

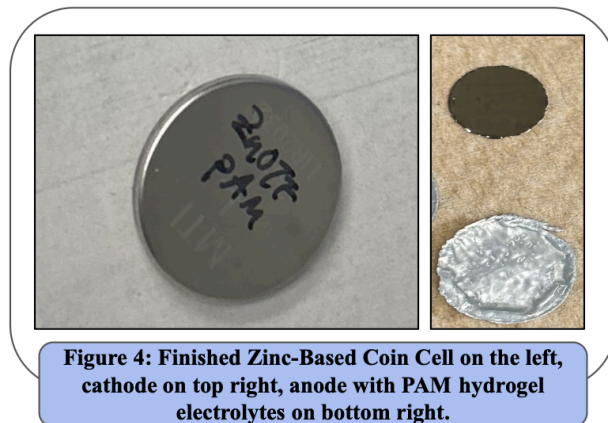
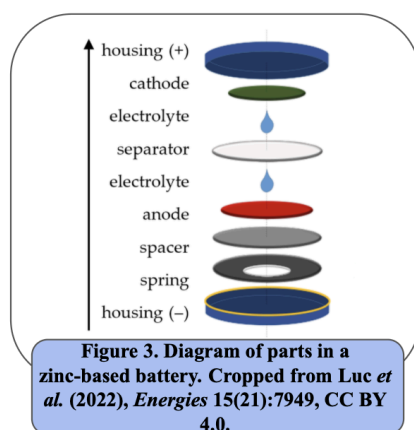
This project must target every challenge previously stated while considering overall performance pertaining to three traits: energy density, cycle retention, and cost efficiency. Firstly, the negative environmental impact of lithium extraction can be solved by using zinc, a more common and eco-friendly material, rather than lithium. Next, to suppress severe dendrite growth, this project uses gel polymer electrolytes, which are mechanically resilient and prevent the dendrite structures from piercing through any parts. Also, using gel polymer electrolytes also solves for toxic leakage, as the gel-like material is physically unable to leak in contrast to the liquid electrolytes. Furthermore, the gel polymer electrolytes are non-flammable, making it less likely for the battery to set on fire. Finally, this project makes the battery especially resilient as a cell-pouch is used instead of a solid case. This design enables mechanical flexibility, which is crucial for applications operating in harsh or demanding environments.

After implementing all of these modifications, this project measures the energy density, cycle retention, and cost between the standard lithium-ion battery and the novel zinc-based battery. In comparison to other previous studies, this project brings together an overall analysis and solves for many issues instead of just one and compares the standard to the prototype. The prototype is based upon some other studies, including the gel polymer electrolyte recipe. By evaluating these factors collectively rather than in isolation, the project aims to assess the overall value of zinc-based gel polymer electrolyte batteries relative to standardized lithium-ion batteries.

Approach

In this project, two types of batteries will be used, the zinc-based gel polymer electrolyte coin cell and cell pouch. The coin cell will be used to compare the two types of battery (lithium and zinc) to check for energy density, cost, and cycle retention. The cell pouch would be to prove the ability of mechanical resilience and flexibility in the battery and will be the official prototype.

In order to fabricate the zinc-based battery, we use a coin cell as the frame to hold the parts. We polish the metal zinc anode in order to get rid of impurities. Next, we take the electrolytes and put it on the anode using a pipette and carefully center the separator on the liquid. When placed on top of the PAM hydrogel electrolytes, make sure that the electrolytes don't spill out. The cathode is made of 70% NaVO Cathode, 20% Carbon Black, and 10% PVDF. Place the cathode on the positive terminal of the coin cell, and then place the separator-electrolyte-anode on top, the separator facing the cathode. Then, put the spacer on the anode, and finally, put the spring on the very top. Close the coin cell with the negative terminal, and then use a press to make sure the coin cell is secure. Wipe away any excess liquids, and then the zinc-based gel polymer electrolyte coin cell is made.



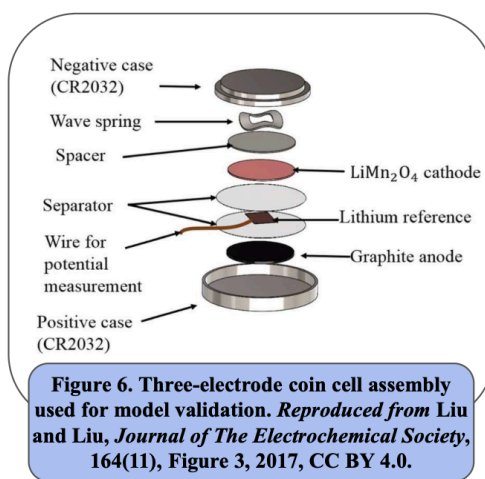
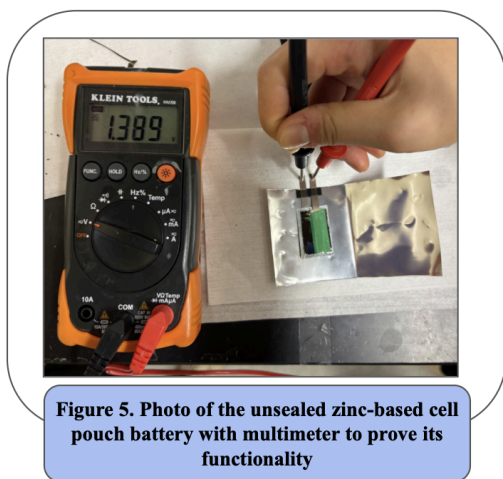
After, in order to make the zinc-based gel polymer electrolyte cell pouch, we would use aluminum in a sheet and cut out two congruent rectangles. Then, we make the PAM

hydrogel electrolyte solution. Carefully use a pipette to cover the anode in the solution and then cover it with the cathode. Wait for the solution to become a gel-like substance, and then seal the pouch (See figure 5 below for photo).

Finally, the last component to make in the project is the lithium-ion coin cell battery to use as control to compare the energy density, cost, and cycle retention with the zinc-based battery. First, we make the slurry for the cathode and anode. The slurry calls for active cathode material, NMP, conductive carbon, and PVDF. Apply the slurry on carbon for the anode and aluminum for the cathode. Put them in the oven to bake and then punch small circles out of each. Take the positive terminal of the coin cell and place the anode on top. Place the lithium foil on top and then add the separator. Add the cathode, and finally, the spacer, wave spring, and the negative terminal of the coin cell (See figure 6 for diagram).

Resources

Materials needed for **zinc-based coin cell**: Coin cells, zinc foil, sandpaper, PAM hydrogel



electrolytes (made in current lab), NaVO Cathode (made in current lab), Carbon Black, PVDF, Separator, Spring, Spacer, multimeter.

Additional materials needed for **zinc-based cell pouch**: aluminum foil, conductive tape, stainless steel foil.

Additional materials needed for the **lithium-ion coin cell**: active cathode material (made in current lab), lithium foil (made in current lab), NMP, conductive carbon, wave spring.

Additional Materials needed for **testing**: wires, scale (provided by current lab), caliper.

Goals and Timeline

The criteria for finishing is when the zinc-based batteries achieve a higher energy density, cycle retention, and cost efficiency than the lithium based while also fixing all of the problems listed previously.

We will test these values by using multimeters and calculating the volume and mass for the energy density. For the cycle retention, we use the battery cyclers analyzer to measure for the values. For the cost efficiency, we keep track of the cost for each of the coin cells.

February: Find optimal recipe for PAM hydrogel electrolyte solution and finish making around 5 zinc-based gel polymer electrolyte batteries and make sure all of them work. Create a budget sheet for each battery.

March: Find recipe for liquid electrolyte and slurry for lithium-ion based batteries and finish making 5 of them and make sure all of them work. Update budget sheet for each battery

April: Finish making around 3 zinc-based gel polymer electrolyte batteries and make sure all of them are fully sealed and working. Update the budget sheet for each battery.

May: Finish measuring energy density, cycle retention, and data for cost. Modify zinc-based batteries if they come out to be worse or lower in any of the three criterias. Create graphs and tables for all meaningful data and start making the final presentation.

June: Finish final presentation and revise.

Risks

Chemical fumes could be dangerous if inhaled or get into eyes. There could also be skin irritation if incorrectly used. In order to prevent this from happening, I will use proper safety equipment, such as gloves, safety glasses, and masks to protect myself. Secondly, if any of the batteries do not work, it is important that we remake them. Thirdly, if the recipe for a polymer electrolyte or a solution doesn't work I would do extra research online and find other recipes, and I will try to test out different solutions to understand and acquire the most efficient one.

Current Progress and Need for Funding

I have read many papers on this topic, and I'm currently wanting to do a project related to this topic, but it's difficult to complete a self-initiated project without guidance and funding. I have built the zinc-based gel polymer coin cell successfully, but I'm having trouble finding how to properly seal the zinc-based cell pouch without destroying the circuit.

In my project, I still need to find out how to properly seal the zinc-based cell pouch without impeding the functionality of the battery. I have not, however, been able to make the lithium-ion battery. I also have to measure the energy density, cycle retention, and cost of the zinc and lithium based batteries. Additionally, repeating trials is crucial because there may be a range of values of energy capacity between the batteries, and I want to get the most accurate data. The funding would also help me buy tools to conduct my experiments.

Project Budget

Name	Cost	Quantity	Total Cost	Link
Aluminum Foil Sheet	\$15	1	\$15	Link
Zinc Foil	\$27.26	1	\$27.26	Link
Coin cell	\$54	1	\$54	Link
Separator	\$38	1	\$38	Link

Wave Spring	\$64.9	1	\$64.9	Link
Lithium Foil	\$119.68	1	\$119.68	Link
Spacer	\$38	1	\$38	Link
Multimeter	\$29.98	1	\$29.98	Link
Caliper	\$12	1	\$12	Link
Stripped Wire	\$5	1	\$5	Link
Sandpaper	\$8.98	1	\$8.98	Link
Stainless Steel Foil	\$43	1	\$43	Link
PVDF	\$66.8	1	\$66.8	Link
Conductive Tape	\$4.03	1	\$4.03	Link
Carbon Black	\$23.5	1	\$23.5	Link
NMP	\$105	1	\$105	Link
Conductive Carbon	\$56	1	\$56	Link
Other Chemicals	\$0	0	\$0	Made in Lab

Personal

During my freshman year, I was able to intern under the aerospace engineering department in Cornell. During my sophomore year, I was also able to attend the Houston Science Fair. I created a project based on repurposing abandoned oil rigs by attaching wave energy converters, a machine that converts ocean waves into electricity. I had become especially concerned with environmental issues during this project. My project eventually won a special award. That summer, I joined a program called the Army Educational Outreach Program, where I was given a stipend of \$4,500 to work with a mentor and create a project. I worked with superhydrophobic trimodal surface structures on PDMS and the prevention of biofilms. In the

first semester of junior year, I was able to attend a conference from the American Physical Society as a co-author for a poster. Currently, I'm working in a lab in the University of North Texas under the material science department to explore the effects of different materials used in batteries. As an aspiring electrical engineer, I found working with batteries and their environmental impact inspiring.

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